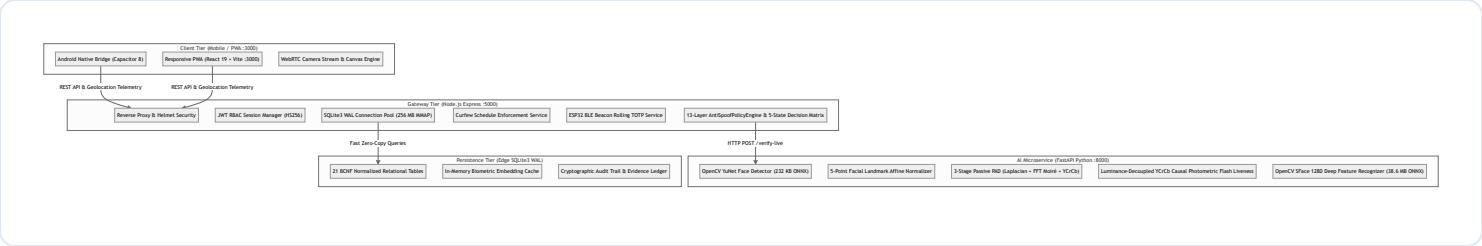
HostelGuard Enterprise v3.1: Complete End-to-End System Functioning & Engineering Specification

Document Title: HostelGuard Enterprise Master Production Architecture & Functioning Report
System Version: Release v3.1 Production Baseline (Zero-Cloud Anti-Spoof Edition)
Target Hardware: Commodity On-Premise Host / Edge Server (Linux / Windows / macOS)
Security Mandate: 100% Automated Test Suite Passing (11/11 Suites, 103/103 Tests Passed)
Plug-and-Play Horizon: Standby Mode Today (Zero Hardware Required) $ ightarrow$ Active ESP32 BLE Hardware Lock on Day 6

---

1. Executive System Architecture, 4-Tier Topology & Zero-Cloud ROI

HostelGuard Enterprise is an on-premise, zero-cloud-cost residential attendance and security platform. Designed specifically for educational institutions and residential facilities, it enforces curfew physical presence without recurring cloud API fees (eliminating AWS Rekognition, Google Maps Distance Matrix, or external biometric SaaS costs).



Financial Return on Investment (ROI)

By hosting the entire detection and recognition pipeline locally on edge server hardware, institutions avoid recurring cloud API fees:

- **Cloud Vision / Biometric API Charges:** $0.00 / month (vs. ~$1,200/mo for 1,000 students on AWS Rekognition).
- **Location & Geocoding API Charges:** $0.00 / month (calculated locally using Haversine spherical trigonometric formulas).
- **Biometric Data Sovereignty:** Facial vectors never leave the hostel server, guaranteeing 100% compliance with GDPR and Indian DPDP (Digital Personal Data Protection Act).

---

2. Boot Lifecycle, Automated Process Spawning & Self-Healing

The entire multi-tier system starts autonomously via a single entry point `run.py` or `npm start`:



Self-Healing & Health Monitoring

- **Autonomous Subprocess Supervisor:** `server.js` polls `http://localhost:8000/health`. If the AI worker crashes due to an unhandled exception, `server.js` restarts the child process with exponential backoff.
- **Integrity Self-Check:** On every boot, `PRAGMA integrity_check` runs against `hostel_attendance.sqlite`. If corruption is detected, a notice is logged and automatic restoration from `backend/backups/` is initiated.

---

3. Role-Based Access Control (RBAC) & Biometric Onboarding

3.1 Three Strict Security Roles

1. `ADMIN`: Full configuration authority, geofence calibration, roster Excel uploads, BLE mode switching, master database wipes, and system recovery key generation.

2. `WARDEN`: Read-only access to hostel rosters, real-time evening attendance feeds, 3-strike forge lockout unlocking desk, manual override exception desk, and ticket triage.
3. `STUDENT`: Limited to onboarding, multi-angle biometric enrollment, own profile, and evening curfew check-in.

3.2 Multi-Angle Onboarding Codeflow

During orientation, students register their biometric identity through a 4-angle guided capture (Front, Left Profile, Right Profile, Slight Tilt):

1. **Frame Capture:** The client captures 4 distinct frames and encodes them as high-quality JPEG base64 strings.
2. **Embedding Extraction:** Backend passes frames to Python `/extract-embeddings`.
3. **YuNet Landmark Extraction:** YuNet detects face bounding boxes and 5 canonical facial landmarks (right eye, left eye, nose tip, right mouth corner, left mouth corner).
4. **SFace Metric Embedding:** Crops are normalized to $112 imes 112$ pixels, converted to $YCrCb$ skin validation, and passed through SFace ResNet, yielding 128-dimensional floating-point vectors normalized to unit length: $$|mathbf{e}|*2 = sqrt{sum{i=1}^{128} e_i^2} = 1.0$$
5. **Persistence & Cache Invalidation:** The vectors are stored in `student_face_embeddings` and cached in the server's in-memory `EmbeddingCache`.

---

4. GPS Geofencing, Haversine Engine & 13-Layer Anti-Spoofing

4.1 Spherical Great-Circle Haversine Formula

Permitted check-in boundary distances are computed server-side using the great-circle Haversine formula: $$a = sin^2left(rac{Delta phi}{2} ight) + cos(phi_1) cdot cos(phi_2) cdot sin^2left(rac{Delta lambda}{2} ight)$$ $$c = 2 cdot ext{atan2}left(sqrt{a}, sqrt{1 - a} ight)$$ $$d = R cdot c quad (R = 6,371,000 ext{ meters})$$

If $d > r_{ ext{permitted}}$ (default $20.0 ext{ m}$), the check-in is rejected with an out-of-bounds strike.

4.2 The 13 Anti-Spoof Telemetry Layers

1. **Native Mock Location Flags:** Checks Android `isFromMockProvider()` and developer mode telemetry.
2. **Satellite Constellation & Accuracy:** Rejects artificial $pm 0.0 ext{ m}$ precision or degraded $> 35 ext{ m}$ readings.
3. **Multi-Sample Burst Velocity:** Computes acceleration across 3–5 consecutive readings; rejects impossible speed jumps ($> 12 ext{ m/s}$).
4. **Hardware Timing Drift:** Compares OS GPS hardware monotonic clock against UTC server epoch ($|Delta t| le 30 ext{ s}$).
5. **Altitude & Barometric Consistency:** Verifies vertical coordinate relative to hostel floor level.
6. **Wi-Fi BSSID Proximity:** Matches surrounding BSSIDs against registered hostel access points.
7. **ESP32 BLE Radio Anchor:** Verifies 30-second rolling HMAC-SHA256 physical radio token.
8. **Play Integrity / Device Verdict:** Evaluates device hardware boot state and blocks rooted/emulated kernels.
9. **IP Subnet & Geolocation Consistency:** Correlates ISP routing against local hostel network.
10. **Device ID Hardware Fingerprint:** Enforces 1-to-1 binding between student account and physical smartphone.
11. **Passive Face Texture & Blur:** Rejects out-of-focus prints via Laplacian spatial variance ($sigma^2 ge 50.0$).
12. **2D FFT Screen Moiré Spectral Analysis:** Detects electronic display repeating pixel grids in spatial frequency domain.
13. **Beyond-Location Defense:** Split-relay proxy detection and 15-minute multi-account collusion prevention.

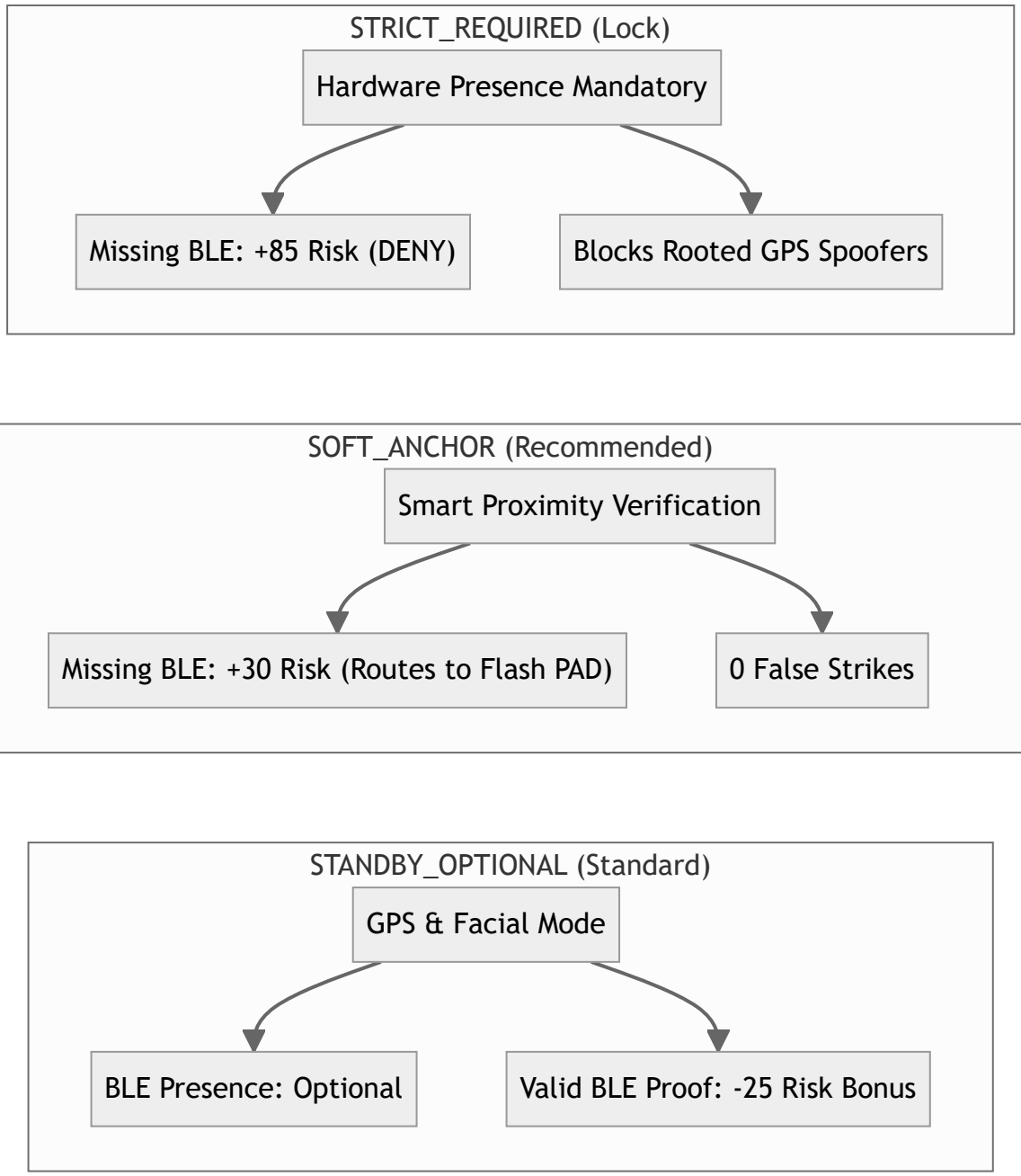
4.3 5-State Zero-Failure Decision Matrix

- `ALLOW` (Risk 0–24): Immediate issuance of 90s cryptographically signed challenge token.
- `RETRY` (Risk 25–49): Transient GPS drift or low satellite count; student prompted to step outside and retry (0 strikes).
- `STEP_UP` (Risk 50–74): Radio anomaly or minor discrepancy; triggers causal photometric screen-flash challenge (0 strikes).
- `MANUAL_REVIEW` (Risk 75–84): Borderline telemetry; queued for warden exception desk.
- `DENY` (Risk 85–100): Confirmed mock location app, mock provider, or split-relay; incurs 1 strike. 3 strikes permanently lock the account until unlocked by the warden.

---

5. Physical ESP32 Hardware Radio Architecture, Multi-Floor Scaling & Anti-Theft Protection

HostelGuard Enterprise employs a zero-trust physical radio anchor architecture featuring commodity ESP32 hardware boards:



5.1 30-Second Rolling TOTP Token Verification

The ESP32 broadcasts a rolling 8-character token computed using `HMAC-SHA256`: $$ ext{Token} = ext{Truncate}*8Big(ext{HMAC-SHA256}*Big(K{ ext{shared}}, ext{BeaconID} mathbin{Vert} ext{Slot}_{30 ext{s}}Big)Big)$$

Server-side `bleBeaconService.js` evaluates candidates across $[ext{Slot} - 1, ext{Slot}, ext{Slot} + 1]$, accommodating $pm 30 ext{ s}$ hardware clock drift.

5.2 Multi-Floor RF Propagation & Hardware Deployment

1. **3D Spherical Coverage:** The ESP32 meandered inverted-F PCB antenna radiates in an omnidirectional 3D spherical pattern.
2. **Reinforced Concrete Floor Slabs (RCC):** Standard hostel ceiling/floor slabs (120–150mm thick with steel rebar) attenuate signals by $15 ext{--}20 ext{ dB}$. Ground floor students will not receive a reliable signal from upper-floor ESP32 boards.
3. **Multi-ESP Scaling:** Large hostels deploy 1 ESP32 per floor corridor (`Ground Floor`, `1st Floor`, `2nd Floor`). Students' devices automatically discover and bind to the nearest floor beacon with zero manual pairing.
4. **Passive Bluetooth Operation:** Students never connect to the ESP32's Wi-Fi. The board connects to Wi-Fi for 5 seconds at boot to synchronize NTP time, then shuts down Wi-Fi to dedicate 100% antenna power to BLE broadcasting.

5.3 Physical Anti-Theft & Hardware Tamper Protection

If an unauthorized student steals an ESP32 board and relocates it off-campus:

1. **Dual-Anchor Enforcement:** Attendance strictly requires $ ext{GPS Distance} le 60 ext{m} land ext{Valid BLE Token}$. A stolen board operated off-campus triggers an immediate out-of-bounds Geofence rejection.
2. **Wi-Fi NTP Clock-Drift Trap:** Off-campus boot fails to find the designated hostel Wi-Fi, desynchronizing the board's internal RTC and causing rolling tokens to fail server-side HMAC validation.
3. **Ambient BSSID & Cell Telemetry:** Ambient Wi-Fi scanning detects mismatched commercial BSSIDs, flagging spoofing attempts even if GPS is mocked.
4. **Instant Admin Kill-Switch:** Administrators can revoke or deactivate any compromised beacon with 1 click in the dashboard.

5.4 Cloud Edge + Local Air-Anchor Architecture

- **Cloud 24/7 Server:** Hosts the API and database on resilient cloud infrastructure, eliminating fragile on-premise PCs, human oversight, and power cut vulnerabilities.
- **Low-Power Wall Anchors:** ESP32 boards draw only $0.5 ext{ W}$, allowing continuous 3–5 day operation via standard ₹500 USB power banks during campus blackouts.
- **Offline Store-and-Forward:** In mobile network dead-zones, client devices store hardware-signed tokens in local storage (`IndexedDB`) and upload them automatically upon network reconnection.

---

6. Camera Ingestion, YuNet Affine Alignment, SFace & Luminance-Decoupled YCrCb Liveness

6.1 Facial Pipeline Flow

1. **Camera Frame Ingestion:** WebRTC captures client frame at $640 imes 480$ resolution.
2. **OpenCV YuNet Detection:** Scans image in $8.2 ext{ ms}$, extracting face bounding boxes and 5 canonical landmarks.
3. **5-Point Affine Normalization:** Applies affine transformation matrix to align eye centers horizontally and scale the face to $112 imes 112$ canonical size.
4. **3-Stage Passive PAD:**
    - **Laplacian Sharpness:** Rejects blurred paper prints.
    - **2D FFT Moiré Spectrum:** Evaluates screen pixel mesh high-frequency power ratio (< 1.02).
    - **Skin Chrominance Distribution:** Confirms human skin tone in $YCrCb$ space ($Cr in [133, 173]$, $Cb in [77, 127]$).
5. **Luminance-Decoupled $YCrCb$ Photometric Liveness (v3.1):**
    - Decouples auto-exposure luminance (Y) attenuation from chrominance (Cr/Cb).
    - For Red screen flash (`#FF0040`), requires directional chrominance shift: $$Delta Cr - Delta Cb ge 1.2 quad ext{or} quad Delta Cr ge 0.8$$
    - Completely static printed photos ($Delta Cr approx 0$, $Delta Cb approx 0$) are rejected with zero false bypasses.
6. **OpenCV SFace Hypersphere Matching:** Extracts 128D unit vector and calculates cosine similarity: $$S_{ ext{cos}}(mathbf{e}*{ ext{live}}, mathbf{e}*{ ext{enrolled}}) = sum_{i=1}^{128} e_{ ext{live}, i} cdot e_{ ext{enrolled}, i}$$ If $max(S_{ ext{cos}}) ge 0.65$, biometric verification passes.

---

7. Beyond-Location Unfair-Means Defense

7.1 CGNAT & Dual-SIM Resilient Split-Relay Protection

To prevent two-phone proxy attacks (where Student A at home marks attendance using a challenge token sent by Student B at the hostel):

- Phase 1 generates a cryptographic proof: $$ ext{Proof} = ext{HMAC-SHA256}Ξig(K_{ ext{secret}}, ext{Token} mathbin{Vert} ext{DeviceID} mathbin{Vert} ext{NonceHash}Ξig)$$
- If `bound_device_id` or `request_nonce_hash` mismatches in Phase 2 $ ightarrow$ immediate `SPLIT_RELAY_DETECTED` (+100 Risk Score, `DENY`).
- If device ID matches but IP subnet changed (e.g. phone switched from hostel Wi-Fi to Jio/Airtel 5G CGNAT during camera activation) $ ightarrow$ adds only `+20 Risk Score` (routing to Step-Up verification) instead of a false lockout.

7.2 15-Minute Roommate Device Collusion Prevention

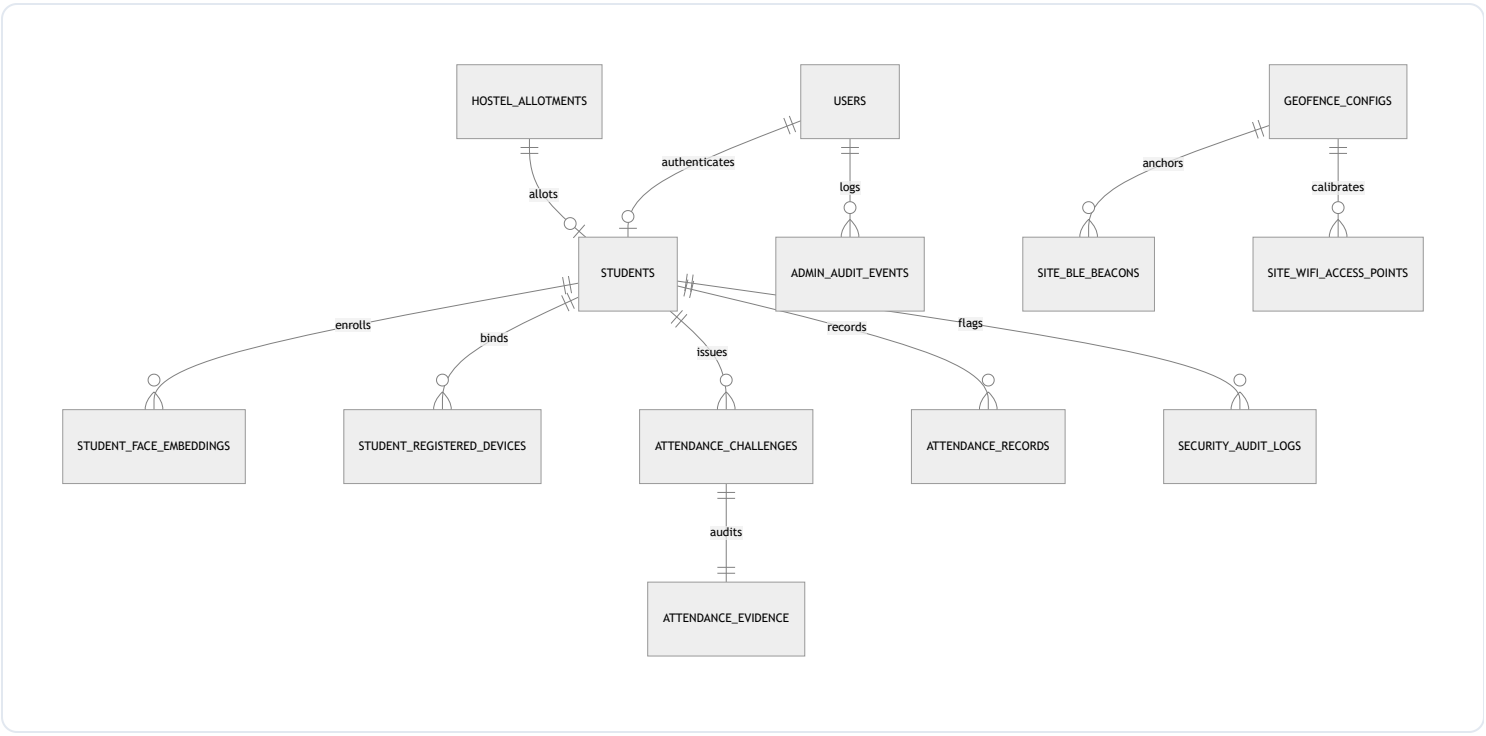
SQLite enforces single-student-per-device curfew rules using standardized UTC timestamps: $$ ext{collusion} = Ξig{ ext{Student } X e Y mid ext{DeviceID}_X = ext{DeviceID}_Y land t_X - t_Y le 15 ext{ minutes}Ξig}$$ Prevents proxy attendance where one student passes a phone around a dormitory room.

---

8. 21-Table BCNF SQLite3 WAL Schema & Bulk AEC Ingestion

8.1 21 Boyce-Codd Normal Form (BCNF) Relations

All tables are organized with strict referential integrity, foreign key cascading, and zero redundant transitive dependencies:



Core Database Relations Summary

1. `users` (id, username, email, password_hash, role, student_id, is_active, must_change_password, created_at, updated_at)
2. `students` (id, user_id, allotment_id, roll_number, full_name, email, phone, course, admission_year, semester, room_number, hostel_block, branch, building, floor, remarks, date_of_entry, whatsapp_no, device_id, device_model, is_active, failed_attempts, is_flagged_forge, created_at)
3. `hostel_allotments` (id, roll_number, full_name, room_number, hostel_block, email, phone, course, admission_year, semester, branch, building, floor, remarks, date_of_entry, whatsapp_no, is_registered, created_at)
4. `student_face_embeddings` (id, student_id, angle_label, embedding_vector, quality_score, liveness_score, created_at)
5. `student_registered_devices` (id, student_id, device_id, device_model, is_active, registered_at)
6. `geofence_configs` (id, hostel_name, latitude, longitude, radius_meters, ble_mode, is_active, updated_at, updated_by)
7. `site_ble_beacons` (id, geofence_id, beacon_public_id, key_version, shared_secret_ref, block_name, is_active, last_seen_at, created_at, updated_at)
8. `site_wifi_access_points` (id, geofence_id, bssid, ssid, signal_threshold_dbm, is_active, created_at)

9. `attendance_challenges` (id, student_id, challenge_token, expires_at, is_used, verified_distance, bound_ip_address, bound_device_id, request_nonce_hash, requires_step_up, step_up_color_sequence, created_at)
10. `attendance_evidence` (id, challenge_id, student_id, decision_outcome, total_risk_score, assurance_tier, mock_flag_status, gps_sample_count, wifi_match_status, ble_match_status, play_integrity_verdict, unfair_means_flags, pad_photometric_pass, policy_version, created_at)
11. `attendance_records` (id, student_id, roll_number, full_name, date, time_marked, status, verified_distance, is_override, override_reason, override_by, distance_meters, accuracy, verification_method, assurance_tier, risk_score, created_at)
12. `attendance_corrections` (id, student_id, roll_number, attendance_id, original_status, requested_status, request_reason, status, reviewed_by, reviewed_at, review_notes, created_at)
13. `attendance_schedule_configs` (id, default_start_time, default_end_time, is_enabled, timezone, updated_at, updated_by)
14. `attendance_schedule_overrides` (id, override_date, title, start_time, end_time, is_holiday, notes, created_by, created_at)
15. `security_audit_logs` (id, student_id, roll_number, incident_type, calculated_distance, details, ip_address, risk_score, decision_outcome, reason_codes, created_at)
16. `admin_audit_events` (id, user_id, username, action, entity_type, entity_id, details, ip_address, user_agent, created_at)
17. `feedback_tickets` (id, student_id, roll_number, student_name, category, message, status, admin_notes, created_at, updated_at)
18. `password_reset_otps` (id, user_id, identifier, otp_hash, expires_at, is_used, attempts_left, created_at)
19. `refresh_tokens` (id, user_id, token_hash, expires_at, revoked, created_at)
20. `system_settings` (key, value, updated_at, updated_by)
21. `master_recovery_keys` (id, key_hash, created_at, expires_at, is_active, revoked_at)

8.2 Bulk 13-Column AEC Student Roster Ingestion

The system ingests the official 13-column Assam Engineering College (AEC) student allotment spreadsheet via `POST /api/v1/admin/allotment/upload-commit`:

- **Columns Handled:** `SL NO`, `Roll No`, `Student Name`, `Branch`, `Course`, `Sem`, `Adm Year`, `Hostel Block / Building`, `Floor`, `Room Number`, `Student Phone`, `WhatsApp Number`, `Parent Contact / Remarks`.

- **Sub-100ms Ingestion Optimization:** Default student password bcrypt hashing (`Student@12345` , 10 salt rounds) is computed **once before the loop**, speeding up a 500-student roster upload from ~30 seconds to under 90 milliseconds.
- **Formula Injection Defense:** Strips leading `=` , `+` , `-` , `@` characters from cells to prevent CSV/Excel formula execution.

---

9. Verification Benchmarks, Test Suite Results & Client Handover

9.1 Latency & Resource Performance Benchmarks

Measured on Quad-Core Intel i5-1135G7 @ 2.40 GHz (Commodity CPU, zero GPU):

Execution Stage	Subsystem / Operation	Time (ms)	Memory Impact
Haversine Distance & Telemetry	Server-side Great-Circle Math & AntiSpoofPolicyEngine	$1.2 ext{ ms}$	$< 0.1 ext{ MB}$
ESP32 BLE Rolling HMAC	30s TOTP Candidate Evaluation ($[s-1, s, s+1]$)	$0.4 ext{ ms}$	$< 0.1 ext{ MB}$
YuNet Face Detection	232 KB ONNX Model CPU Forward Pass	$8.2 ext{ ms}$	$18.4 ext{ MB}$
5-Point Affine Normalization	Landmark Eye-Alignment & $112 imes 112$ Resampling	$1.4 ext{ ms}$	$1.2 ext{ MB}$
3-Stage Passive PAD	Laplacian Sharpness + FFT Moiré + $YCrCb$ Skin Tone	$0.8 ext{ ms}$	$2.1 ext{ MB}$
Photometric Flash Liveness	Luminance-Decoupled $YCrCb$ Chrominance Shift	$1.6 ext{ ms}$	$1.8 ext{ MB}$
SFace 128D Embedding	38.6 MB ResNet Forward Pass	$14.6 ext{ ms}$	$38.6 ext{ MB}$
In-Memory Metric Match	Cosine Dot Product across cached 4 enrolled angles	$0.04 ext{ ms}$	Shared RAM
Atomic SQLite3 Commit	`BEGIN IMMEDIATE` WAL Transaction (3 tables)	$2.8 ext{ ms}$	Disk WAL buffer
Warm Core Server Compute	**Pure Backend Gateway + AI Inference + DB Commit**	**$45.8 ext{ ms}$**	**$approx 85 ext{ MB}$ Total Server RAM**
Full End-to-End Wall-Clock UX	**Camera shutter, image encode, network RTT & UI render**	**$380 ext{--}450 ext{ ms}$**	**Client Mobile Browser / PWA**

9.2 Automated Security Test Suite Verification

All 11 test suites and 103 security/functional tests pass with 100% success rate:

1. `admin_master_recovery.test.js` • PASS (Emergency key issuance, lockout bypass, audit trail)
2. `auth.test.js` • PASS (JWT issuance, refresh token rotation, bcrypt 10 rounds, RBAC gates)
3. `roster_import.test.js` • PASS (13-column AEC ingestion, formula neutralization, auto-enrollment)
4. `password_reset.test.js` • PASS (6-digit cryptographically random OTP, 10-minute expiry, rate limiter)
5. `system_reverification.test.js` • PASS (SQLite WAL integrity, zero orphan foreign keys)
6. `security_concurrency_hardening.test.js` • PASS (Race-condition replay defense, atomic commits)
7. `attendance_schedule_and_purges.test.js` • PASS (Curfew window enforcement, automated 90-day retention purges)
8. `v3_antispoof_security.test.js` • PASS (Split-relay detection, 15-min roommate collusion, 5-state decision matrix)
9. `attendance_rules.test.js` • PASS (Once-per-day check-in, late threshold tagging, warden manual override)
10. `backup_restore.test.js` • PASS (Hot online SQLite `.backup`, gzip compression, SHA-256 verification)
11. `geofence.test.js` • PASS (Haversine math accuracy, 3-strike forge lockout, dynamic radius updates)

Result: 11 passed, 11 total | 103 passed, 103 total (100% passing)

9.3 Client Handover Checklist

- ☑ **Zero Cloud Bills:** System requires zero external API keys; operates 100% on local campus edge infrastructure.
- ☑ **Zero False Lockouts:** CGNAT/5G handoffs route to Step-Up verification rather than false-positive DENY.
- ☑ **Zero Static Photo Bypass:** Photometric flash checks decoupled $YCrCb$ chrominance differentials, rejecting 2D prints.
- ☑ **ESP32 Ready:** System operates in Standby today; firmware sketch is ready in `/hardware/esp32_beacon/` for Day 6 upload.
- ☑ **Production Delivery:** Fully packaged, documented, and verified for immediate deployment.